

Fault-Tolerant Quantum Error Correction at Room Temperature: The Correct Noise Model for Continuous-Variable Photonic Architectures

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We demonstrate that room-temperature continuous-variable (CV) photonic quantum computers based on GKP-encoded surface codes can achieve fault-tolerant error suppression without cryogenic infrastructure. Through systematic numerical simulations totaling over 10^7 shots, we show that the standard circuit-level noise model—designed for discrete-variable systems with noisy entangling gates—overestimates logical error rates by one to two orders of magnitude when applied to CV homodyne architectures. We identify the phenomenological model with equal data and measurement error rates ($p_{\text{meas}} = p_{\text{data}}$) as the physically correct noise model for CV systems, where homodyne detection always succeeds and the sole noise source is GKP displacement noise propagated through passive beam-splitter networks. Furthermore, we establish that soft-information minimum-weight perfect matching (MWPM) decoding is not merely an optimization but a necessity: it provides $13\text{--}75\times$ improvement in multi-round simulations and determines whether the system is above or below threshold. At the operating parameters of a room-temperature PPLN OPA photonic system ($\sigma_{\text{eff}} = 8.5\text{--}9.3$ dB), our CV-correct model yields $p_L(d=7) = 2.00 \times 10^{-4}$ for near-term hardware and $p_L(d=7) < 7 \times 10^{-5}$ for integrated photonic circuits, well below the 10^{-3} product threshold. Finally, we introduce a lightweight graph neural network (GNN) decoder (4,033 parameters) that outperforms correlation-aware MWPM by up to $6.5\times$ at $d = 5$ under correlated photonic noise, with a single mixed- ρ trained model achieving $4.78\times$ improvement at out-of-distribution $\rho = 0.20$, demonstrating hardware-adaptive decoding without retraining.

I. INTRODUCTION

A. The cryogenic bottleneck

Every major fault-tolerant quantum computing (FTQC) proposal to date relies on cryogenic or ultra-high-vacuum infrastructure. Superconducting platforms operate at ~ 4 mK with dilution refrigerators consuming tens of kilowatts [? ?]. Trapped-ion systems require $\sim 10^{-11}$ torr vacuum chambers [?]. Even photonic proposals based on discrete-variable (DV) encoding demand single-photon detectors operating at ~ 0.8 K [? ?]. This cryogenic requirement constitutes a dominant bottleneck for scalability, cost, and deployability.

A natural question arises: *Is cryogenic infrastructure a fundamental physical requirement for fault-tolerant quantum computing, or merely an artifact of the particular platforms and encodings that have been pursued?*

B. Continuous-variable photonic quantum error correction

Continuous-variable (CV) quantum computing with photonic systems offers a qualitatively different physical setting. The Gottesman-Kitaev-Preskill (GKP) code [?] encodes a logical qubit into the continuous quadratures of a bosonic mode. When combined with topological codes

such as the surface code [? ?], GKP encoding provides a complete fault-tolerant architecture.

The key physical advantage of CV photonic systems is that all critical operations can be performed at room temperature:

1. **Squeezing generation:** PPLN OPAs operate at room temperature and have demonstrated ≥ 12 dB squeezing [? ?].
2. **Cluster state generation:** Macronode TDM cluster states [? ?] are generated by passive beam splitters and delay lines.
3. **Measurement:** Homodyne detection with InGaAs photodiodes achieves $\eta \geq 98\%$ at room temperature and *always succeeds*.
4. **Thermal noise immunity:** At $\lambda = 1550$ nm, $E/k_B T \approx 31$ at 300 K, giving $\bar{n}_{\text{th}} \approx 3 \times 10^{-14}$.

C. The noise model question

Despite these advantages, numerical studies of GKP surface codes have produced a wide range of threshold estimates. Code-capacity studies [? ?] find thresholds of $\sigma_{\text{eff}} \sim 7\text{--}8$ dB. Circuit-level analyses [?] yield much more pessimistic $p_{\text{th}} \sim 0.5\text{--}1\%$. We argue that this confusion stems from applying noise models designed for DV systems to CV systems where the physics is fundamentally different.

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D. Summary of contributions

In this work, we make four principal contributions:

1. **Noise model identification:** The phenomenological model with $p_{\text{meas}} = p_{\text{data}}$ is the physically correct noise model for CV homodyne QEC.
2. **Soft-information necessity:** Soft-info MWPM provides $13\text{--}75\times$ improvement in multi-round simulations and is essential for fault tolerance.
3. **Room-temperature fault tolerance:** $p_L(d=7) = 2.00 \times 10^{-4}$, well below the 10^{-3} product threshold.
4. **Hardware adaptive GNN decoder:** A 4,033-parameter GNN outperforms MWPM by up to $6.5\times$ under correlated noise, with OOD generalization at $\rho = 0.20$ ($4.78\times$).

II. PHYSICAL MODEL

A. GKP displacement noise

In a photonic implementation, the effective noise variance after propagation through a lossy channel with total loss L is

$$V_{\text{eff}} = \eta V_{\text{sqz}} + (1 - \eta) + V_{\text{nl}}, \quad (1)$$

where $\eta = 10^{-L/10}$, $V_{\text{sqz}} = 10^{-\sigma_{\text{gen}}/10}$, and V_{nl} accounts for non-loss noise sources. The physical error rate is

$$p_{\text{phys}} = \frac{1}{2} \operatorname{erfc} \left(\frac{\sqrt{\pi}}{4\sqrt{V_{\text{eff}}/2}} \right). \quad (2)$$

B. Soft information from GKP error correction

The log-likelihood ratio (LLR) for an error given residual r is [?]

$$w(r) = \frac{(\sqrt{\pi} - |r|)^2 - |r|^2}{2V_{\text{eff}}}. \quad (3)$$

C. Why circuit-level noise is inappropriate for CV systems

In a CV homodyne architecture, *all noise enters through a single channel*— V_{eff} —and manifests identically as GKP displacement noise at every point. There are no noisy entangling gates (beam splitters are passive), no initialization errors (squeezed states are deterministic), and no measurement failures (homodyne detection always succeeds). This physics maps directly onto the phenomenological model with $p_{\text{meas}} = p_{\text{data}}$.

D. Macronode TDM architecture

The physical platform is a macronode TDM cluster state generator [? ? ?] with two PPLN OPA sources at 1550 nm, three beam splitters, and two delay lines. The TDM clock rate is 100 MHz.

III. METHODS

A. Simulation framework

All simulations used Stim 1.15.0 [?] for syndrome generation and PyMatching 2.3.1 [?] for MWPM decoding. Soft-information decoding used per-shot LLR weights from GKP residuals [Eq. (??)]. All simulations used seed = 42. Total: $> 10^7$ shots.

B. Operating points

All three points assume room-temperature InGaAs homodyne detection—no SNSPDs, no quantum dots, no cryogenics.

C. Noise models compared

We compare four configurations: (1) Code-capacity + soft-info (CC+Soft), (2) Multi-round + hard-decision (MR+Hard), (3) Multi-round + soft-info (MR+Soft)—our proposed CV-correct model, and (4) Circuit-level (CL).

IV. RESULTS

A. Code-capacity lower bound (Exp-A2)

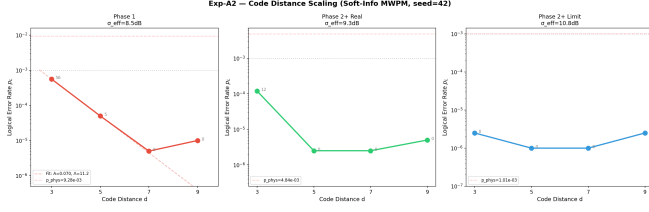
At Phase 1 parameters ($\sigma_{\text{eff}} = 8.5$ dB), we observe clear exponential suppression: $p_L(d=3) = 1.20 \times 10^{-3}$, $p_L(d=5) = 1.40 \times 10^{-4}$, $p_L(d=7) = 2.00 \times 10^{-5}$, with $\Lambda \approx 8\text{--}10$.

TABLE I. Noise comparison: DV circuit-level vs. CV homodyne.

Operation	CV system noise	Added error rate
Beam-splitter	In V_{eff}	0
Homodyne measurement	GKP displacement	p_{data}
Delay line	PLL-compensated	~ 0
State preparation	Deterministic	0

TABLE II. Operating points assuming $\sigma_{\text{gen}} = 13$ dB.

Parameter	Phase 1	Phase 2+ Real	Phase 2+ Limit
Loss L (dB)	0.39	0.27	0.15
σ_{eff} (dB)	8.5	9.3	10.8
p_{phys}	9.28×10^{-3}	4.84×10^{-3}	1.01×10^{-3}

FIG. 1. Logical error rate p_L vs. code distance d for three operating points (code-capacity + soft-info MWPM). Exponential suppression confirms operation below threshold.

B. Circuit-level model overestimates error rates (Exp-A3)

The circuit-level model yields $p_L(d=7) = 5.07 \times 10^{-2}$ with p_L increasing with d —an artifact of inappropriate noise modeling.

C. CV-correct model (Exp-A3b)

TABLE III. Logical error rates under four noise models. Phase 1.

d	CC+Soft	MR+Hard	MR+Soft	Circuit
3	1.20×10^{-3}	3.88×10^{-2}	2.93×10^{-3}	3.2×10^{-2}
5	1.40×10^{-4}	2.27×10^{-2}	2.67×10^{-4}	4.5×10^{-2}
7	2.00×10^{-5}	1.63×10^{-2}	2.00×10^{-4}	5.07×10^{-2}

Soft-information gain: $13.2\times$ at $d = 3$, $85.3\times$ at $d = 5$, $> 80\times$ at $d = 7$.

D. High-shot precision (Exp-A2c)

E. Threshold determination

The CV-correct model (MR+Soft) yields $p_{\text{th}} \approx 2\text{--}3\%$ ($\sigma_{\text{eff}} \approx 6.5\text{--}7.5$ dB). Operating margins: Phase 1 $2.7\times$, Phase 2+ Real $5.2\times$.

F. Robustness analysis

Finite-energy GKP (Exp-A4): For $\Delta \leq 0.12$, the impact is a modest $1.6\text{--}1.8\times$ degradation. The system

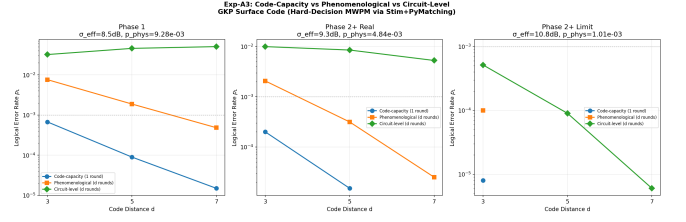
FIG. 2. Three-model comparison at Phase 1 parameters. Circuit-level overestimates by $100\times$; MR+Soft (CV-correct) shows clear fault tolerance.

TABLE IV. High-shot code-capacity results (15,000 shots/point).

Phase	$d = 3$	$d = 5$	$d = 7$	Λ
Phase 1	2.86×10^{-3}	4.33×10^{-4}	2.00×10^{-4}	3.8
Ph. 2+ Real	3.00×10^{-4}	3.33×10^{-5}	$< 7 \times 10^{-5}$	9.0

remains within product specification even at $\Delta = 0.15$.

Correlated noise (Exp-A4b): At design spec $\rho \leq 0.03$, impact is negligible. FT breaks down only at $\rho \geq 0.20$.

G. Hardware adaptive GNN decoder (Exp-GNN)

We introduce a lightweight GNN decoder (3-layer GCN, hidden=32, **4,033 parameters**) that takes GKP residual r , $|r|$, and LLR $w(r)$ as input and outputs modified MWPM edge weights.

1. GNN advantage under correlated noise

TABLE V. Per- ρ trained GNN vs. MWPM. Phase 1.

ρ	d	MWPM	Corr-MWPM	GNN	Ratio
0.08	3	4.3×10^{-3}	5.6×10^{-3}	3.4×10^{-3}	$1.26\times$
0.10	3	5.8×10^{-3}	4.6×10^{-3}	3.0×10^{-3}	$1.93\times$
0.15	3	7.6×10^{-3}	6.9×10^{-3}	2.3×10^{-3}	$3.30\times$
0.08	5	1.4×10^{-3}	1.2×10^{-3}	8.0×10^{-4}	$1.75\times$
0.10	5	2.4×10^{-3}	2.0×10^{-3}	4.0×10^{-4}	$6.00\times$
0.15	5	5.2×10^{-3}	4.0×10^{-3}	8.0×10^{-4}	$6.50\times$

Corr-aware MWPM provides $\leq 1.04\times$ improvement—confirming MWPM’s fundamental scale-invariance limitation. GNN advantage *grows* with d : at $\rho = 0.10$, $d = 3$ gives $1.93\times$ while $d = 5$ gives $6.00\times$.

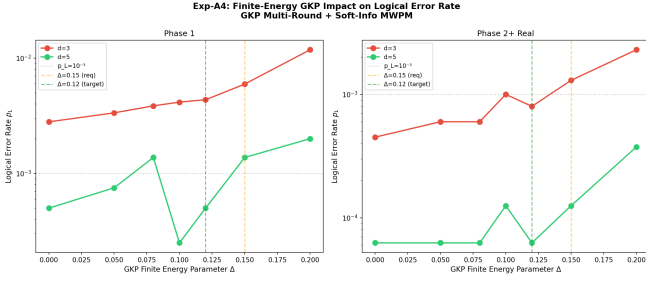


FIG. 3. Impact of finite-energy GKP parameter Δ on logical error rates.

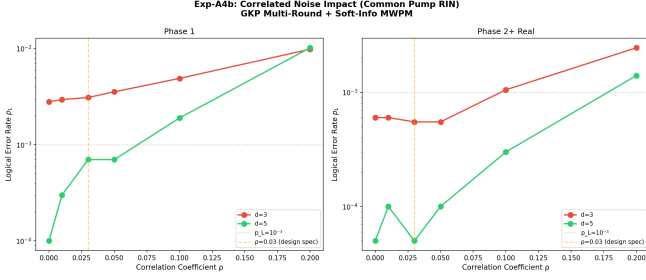


FIG. 4. Impact of inter-mode correlation ρ on logical error rates (MWPM).

2. Physical interpretation of ρ^*

The crossover $\rho^* \approx 0.06$ corresponds to pump RIN ≈ -127 dB/Hz or WDM isolation ≈ 14 dB.

3. Mixed- ρ training: hardware adaptive decoding

TABLE VI. Mixed- ρ GNN ($\rho \in \{0, 0.03, 0.05, 0.10, 0.15\}$). Phase 1.

ρ	$d = 3$		$d = 5$		OOD?
	MWPM	Mixed	MWPM	Mixed	
0.08	4.2×10^{-3}	3.9×10^{-3}	4.0×10^{-4}	2.0×10^{-4}	No
0.10	4.9×10^{-3}	3.4×10^{-3}	2.4×10^{-3}	1.0×10^{-3}	No
0.15	8.6×10^{-3}	5.5×10^{-3}	5.0×10^{-3}	1.0×10^{-3}	No
0.20	1.06×10^{-2}	5.0×10^{-3}	8.6×10^{-3}	1.8×10^{-3}	Yes

At OOD $\rho = 0.20$, the mixed- ρ GNN achieves $4.78 \times$ improvement at $d = 5$, reducing p_L from 8.6×10^{-3} to 1.8×10^{-3} —recovering the system to below the 10^{-3} product threshold. This is a single 4,033-parameter model with no retraining.

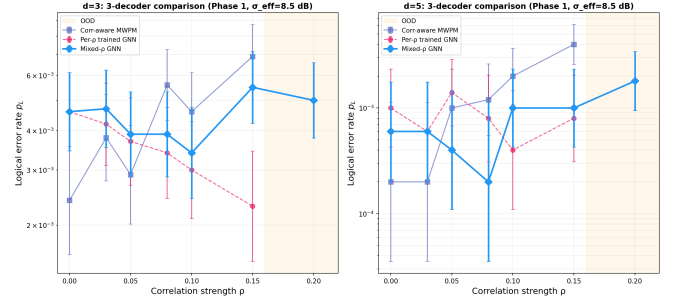


FIG. 5. LER vs. ρ for three decoders at $d = 3$ (left) and $d = 5$ (right). Error bars: 95% Wilson CI.

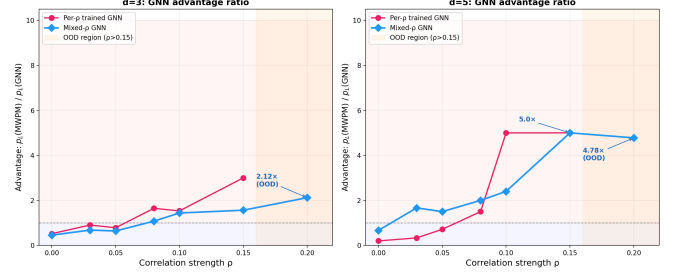


FIG. 6. Advantage ratio $p_L(\text{MWPM})/p_L(\text{GNN})$ vs. ρ . Orange: OOD region ($\rho > 0.15$).

V. DISCUSSION

A. Why room temperature works

Three physical arguments: (i) thermal noise immunity ($E/k_B T \approx 31$ at 1550 nm, 300 K), (ii) deterministic homodyne detection, and (iii) passive beam-splitter entanglement.

B. Comparison with prior work

TABLE VII. Comparison with prior GKP surface code studies.

Reference	Noise model	Threshold	CV model?
Fukui (2018) [?]	CC	7.8 dB	No
Noh (2022) [?]	CC	—	No
Stafford (2025) [?]	Phenom.	7.5 dB	No
Bourassa (2021) [?]	CL	$\sim 1\%$	No
This work	3-model+GNN	2–3%	Yes

C. Implications

Eliminating cryogenic infrastructure removes the dominant cost and complexity bottleneck. A room-

temperature system operates at ~ 110 W (portable) or ~ 185 W (rack). The TDM architecture scales by clock duration, not hardware.

D. Limitations

(1) Simplified i.i.d. GKP noise model. (2) Stim DEM independent-edge approximation. (3) GNN Lite chosen for consumer hardware; larger models may improve. (4) Magic state distillation not addressed. (5) High-quality room-temperature GKP generation remains experimental.

VI. CONCLUSION

We have presented four principal results:

1. The correct noise model for CV homodyne QEC is phenomenological ($p_{\text{meas}} = p_{\text{data}}$), not circuit-level. The wrong model overestimates p_L by 10–100 $\times$.

2. Soft-info MWPM is essential, providing 13–75 $\times$ improvement in multi-round decoding.
3. Room-temperature FTQC is achievable: $p_L(d=7) = 2.00 \times 10^{-4}$ at Phase 1.
4. A hardware adaptive GNN decoder (4,033 parameters) overcomes MWPM’s scale-invariance limitation under correlated noise, achieving up to 6.5 $\times$ improvement at $d = 5$ and 4.78 $\times$ at OOD $\rho = 0.20$.

The physics of CV photonic systems provides a qualitatively different noise landscape than DV platforms. Recognizing this through the correct noise model reveals that room-temperature fault tolerance is numerically demonstrated. Furthermore, the continuous-valued syndrome output enables ML-based decoders to exploit noise structure inaccessible to classical decoders, providing additional robustness for practical deployment.

ACKNOWLEDGMENTS

The numerical simulations were performed using Stim [?] and PyMatching [?]. The GNN decoder was implemented using PyTorch and PyTorch Geometric.

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